A. Cherry

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy | https://codeforces.com/problemset/problem/1554/A

You are given n integers $a_1, a_2, \ldots, a_n$. Find the maximum value of $max(a_l, a_{l+1}, \ldots, a_r) \cdot min(a_l, a_{l+1}, \ldots, a_r)$ over all pairs (l, r) of integers for which $1 \le l$ lt $r \le n$.

Input

The first line contains a single integer t ($1 \le t \le 10\,000$) — the number of test cases.

The first line of each test case contains a single integer n ($2 \le n \le 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^6$).

It is guaranteed that the sum of n over all test cases doesn't exceed $3 \cdot 10^5$.

Output

For each test case, print a single integer — the maximum possible value of the product from the statement.

Example

Input
4 3 2 4 3 4 3 2 3 1 2 69 69 6 719313 273225 402638 473783 804745 323328

Output
12 6 4761 381274500335

Note

Let $f(l, r) = max(a_l, a_{l+1}, \ldots, a_r) \cdot min(a_l, a_{l+1}, \ldots, a_r)$.

In the first test case,

- $f(1, 2) = max(a_1, a_2) \cdot min(a_1, a_2) = max(2, 4) \cdot min(2, 4) = 4 \cdot 2 = 8$.
- $f(1, 3) = max(a_1, a_2, a_3) \cdot min(a_1, a_2, a_3) = max(2, 4, 3) \cdot min(2, 4, 3) = 4 \cdot 2 = 8$.
- $f(2, 3) = max(a_2, a_3) \cdot min(a_2, a_3) = max(4, 3) \cdot min(4, 3) = 4 \cdot 3 = 12$.

So the maximum is $f(2, 3) = 12$.

In the second test case, the maximum is $f(1, 2) = f(1, 3) = f(2, 3) = 6$.